

Classification of Living Things

Lecture-3

TAKE HOME POINTS

- ☐ What is classification
- ☐ Early biological classification
- ☐ Modern biological classification
- ☐ Binomial nomenclature
- ☐ The genus and species concept
- ☐ Why use scientific names
- ☐ The main classification groups (Taxa)
- ☐ Various classification system
- ☐ Phylogenetic system

Classification of Living Things

- ❑ Over **two million** (2,000,000) different kinds of organisms exists
- ❑ **1.5 million** (1,500,000) different kinds have been identified
- ❑ It has been estimated that for each kind of organism now alive, another **400 kinds** once lived but have since become extinct.
- ❑ One billion (1,000,000,000) different kinds of living things may have existed on the earth at one time or another.

❑ Problems

- *How can we keep track of such a bewildering number of organisms?*
- *How can we even name the organisms now alive when no known language has two million words in it?*

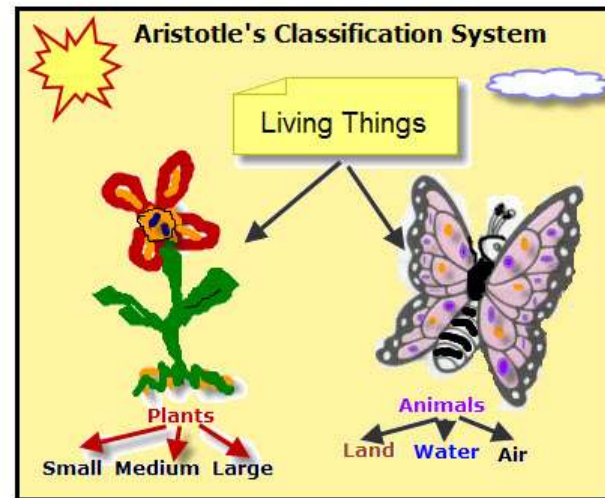
What is Classification?

- Supermarket manager
 - Stamp collectors
 - Word listings in a dictionary
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- *The grouping of similar things for a specific purpose is called **classification***
 - **Early** Biological Classification
 - **Modern** Biological Classification



Early Biological Classification

- How Earliest Humans classify living things
 - Plant Vs Animals or Edible Vs Poisonous Plants or Harmful Vs Harmless animals
- 300 BC, *Aristotle's Classification System*
 - Only about 1000 kinds of organisms were known
 - This classification system survived almost 2000 years

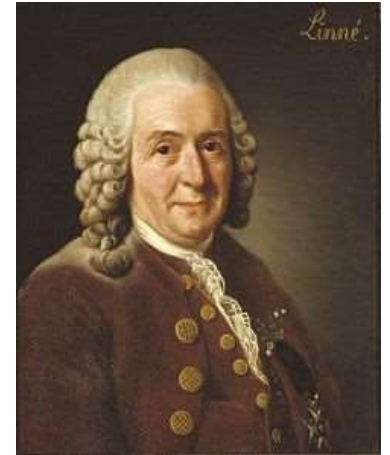


By the beginning of the 18th century, over 10,000 kinds of organisms were known and Aristotle's system was unable to classify them all.

A new system was obviously needed.

Modern Biological Classification

- ***Taxonomy:*** *Taxonomy is the science that deals with the classification of organisms.*
- ***The Contribution of Carolus Linnaeus***
 - ***Diversity:*** *the number of kinds of living things.*
- ***Linnaeus*** grouped organisms according to their **structural similarities**
 - Organisms with very similar structural features were considered to be the same **species**
- Thus all modern-day humans belong to one species, all house cats belong to one species, and all sugar maple trees belong to one species.



Binomial nomenclature

- After classifying, Linnaeus decided to name all the organism.

- Every organism's name is consist of **two words**→ Binomial Nomenclature

- He decided to write all names in *Latin*.

- Example: **Common name:** Human

- **Scientific name:**

Homo sapiens (Italic if printed)

Homo sapiens (separate underlines if handwritten)

Common name: Cat

Scientific name: *Felis domesticus*



Genus and species concept



Genus

- Group of species that are similar.
- Plural is 'genera'.
- Example: Maple trees are from genus *Acer*. So,
- Sugar maple: *Acer saccharum*
- Silver maple: *Acer saccharinum*
- Red maple: *Acer rubrum*
- etc.*

Species

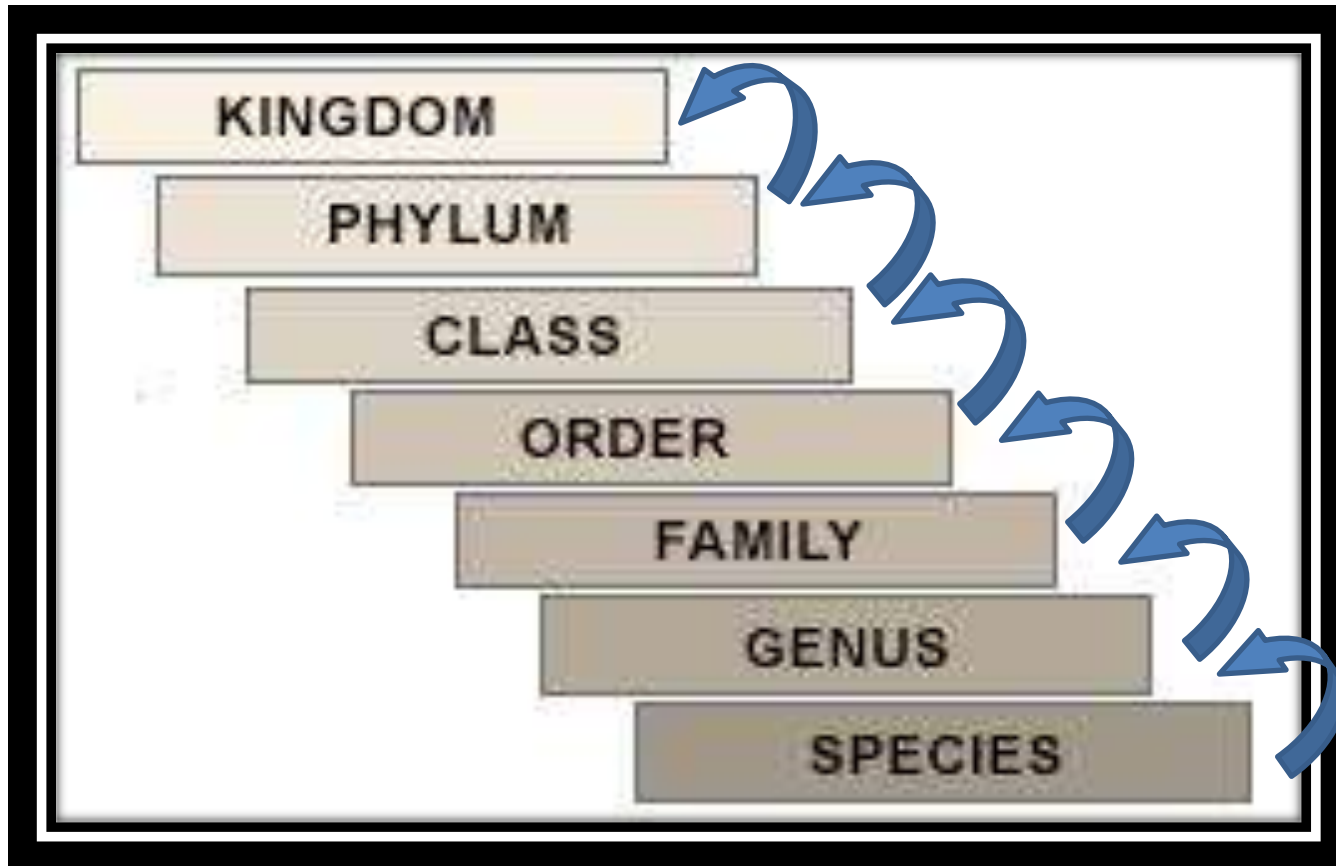
- Organisms that are structurally very similar.
- Plural is also 'species'
- Interbreed under natural condition to produce fertile offspring (children) .
- Example:
Red maple: *Acer rubrum*

Why Use Scientific Names?

- Common names can be confusing or misleading
 - ***Felis concolor*** is called a cougar, mountain lion, puma, panther, painter, and many names
 - Domestic cow is
 - “**la vache**” in French,
 - ‘**die Kuh**’ in German,
 - “**la vaca**” in Spanish, and
 - “**gOrU**” in Bengali
- *Scientific Name: Bos taurus*

Main classification groups(taxa)

There are seven main ***taxa*** or classification groups.



Main classification groups(taxa)

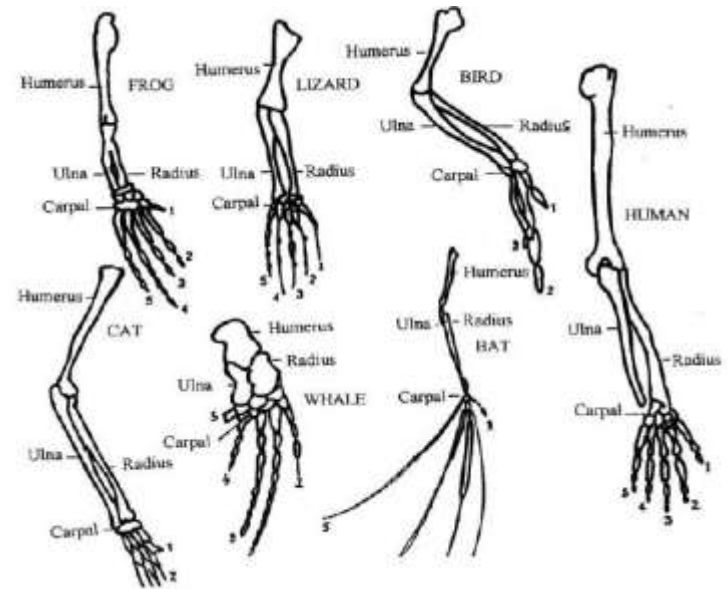
Example:



Taxon	Human	Gorilla
Kingdom	Animalia	Animalia
Phylum or Division	Chordata	Chordata
Class	Mammalia	Mammalia
Order	Primates	Primates
Family	Homonidae	Pongidae
Genus	<i>Homo</i>	<i>Gorilla</i>
Species	<i>sapiens</i>	<i>gorilla</i>

Modern Basis for Classification

- **Homologous Structure**
 - *show the same basic pattern, the same general relationship to other parts, and the same pattern of development.*
- **Similar Biochemistry**
 - *closely related organisms form similar chemical compounds in their body. They use this belief to help classify organisms*
- **Genetic Similarity**
 - *the greater the similarity of DNA among organisms, the more closely they may be related.*

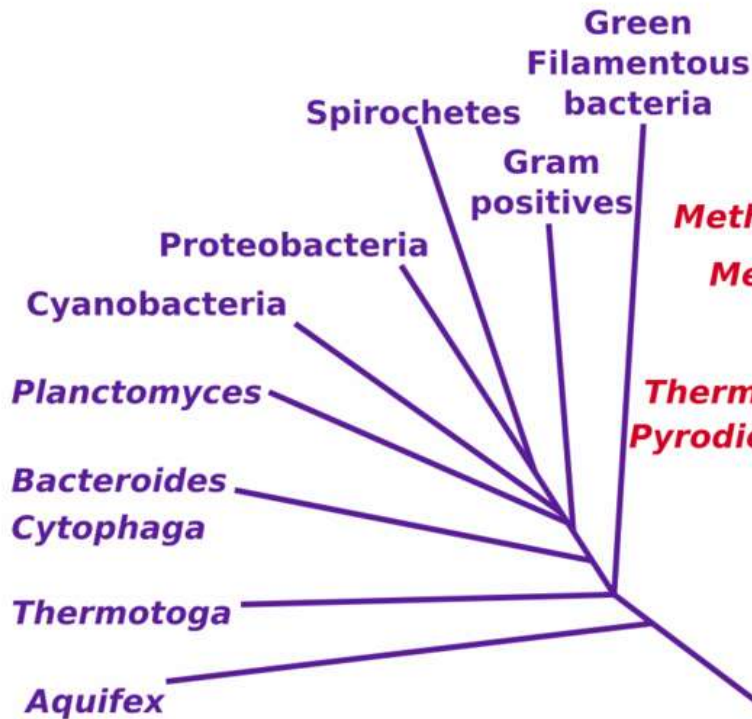


Phylogenetic System

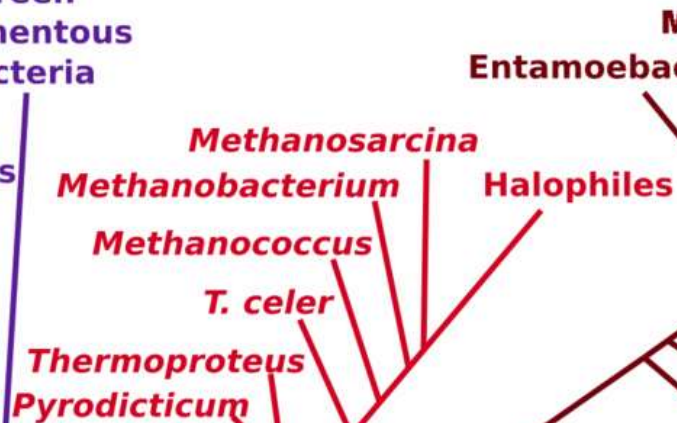
- **Carl R Woese** → completely new approach in classification.
- Used rRNA sequences which are very conserved.
- **Based on Evolution**: Genetic materials (rRNA) were examined and used in revealing evolutionary relationships.
- Organisms are classified into three major **DOMAINS**
 1. Eukarya
 2. Bacteria
 3. Archaea

Phylogenetic Tree of Life

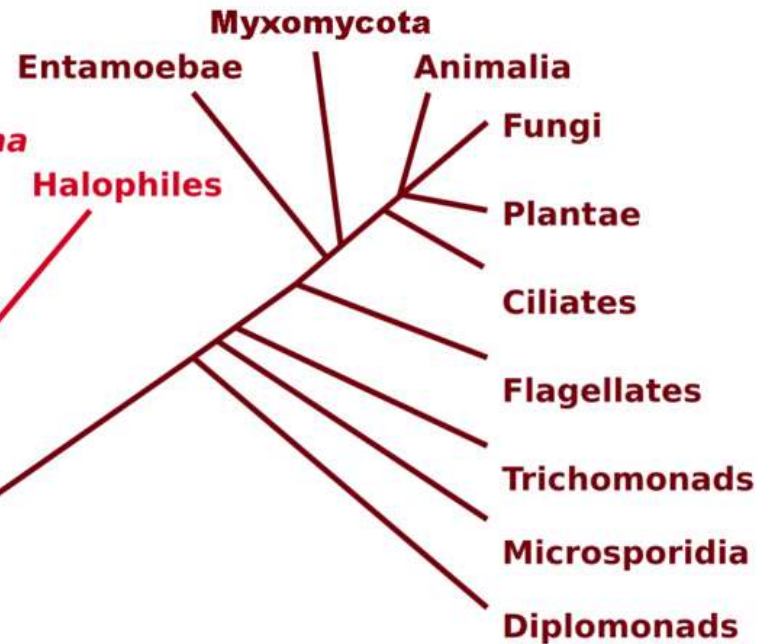
Bacteria



Archaea



Eucarya



Summary of Various classification system

Classification	System Kingdom
2- Kingdom System	1. Plantae 2. Animalia
3- Kingdom System	1. Plantae 2. Animalia 3. Protista
4- Kingdom System	1. Plantae 2. Animalia 3. Protista 4. Monera
5- Kingdom System	1. Plantae 2. Animalia 3. Fungi 4. Protista 5. Monera
6- Phylogenetic system	1. Archaea 2. Bacteria 3. Eukarya

Meet your human taxonomy



KINGDOM

Animalia

(all multicellular organisms that ingest nutrients rather than synthesize them)



PHYLUM

Vertebrata

(all animals with a vertebral column or **dorsal hollow notocord**—a structure along the top of animals—protecting their central nervous system)

Meet your human taxonomy



CLASS

Mammalia

(all vertebrates with placental development, mammary glands, hair or fur, and a tail located behind the anus)



ORDER

Primates

(mammals adapted to life in trees, with opposable thumbs)

Meet your human taxonomy



FAMILY

Hominidae

(primates that move primarily with bipedal—two-footed—locomotion)

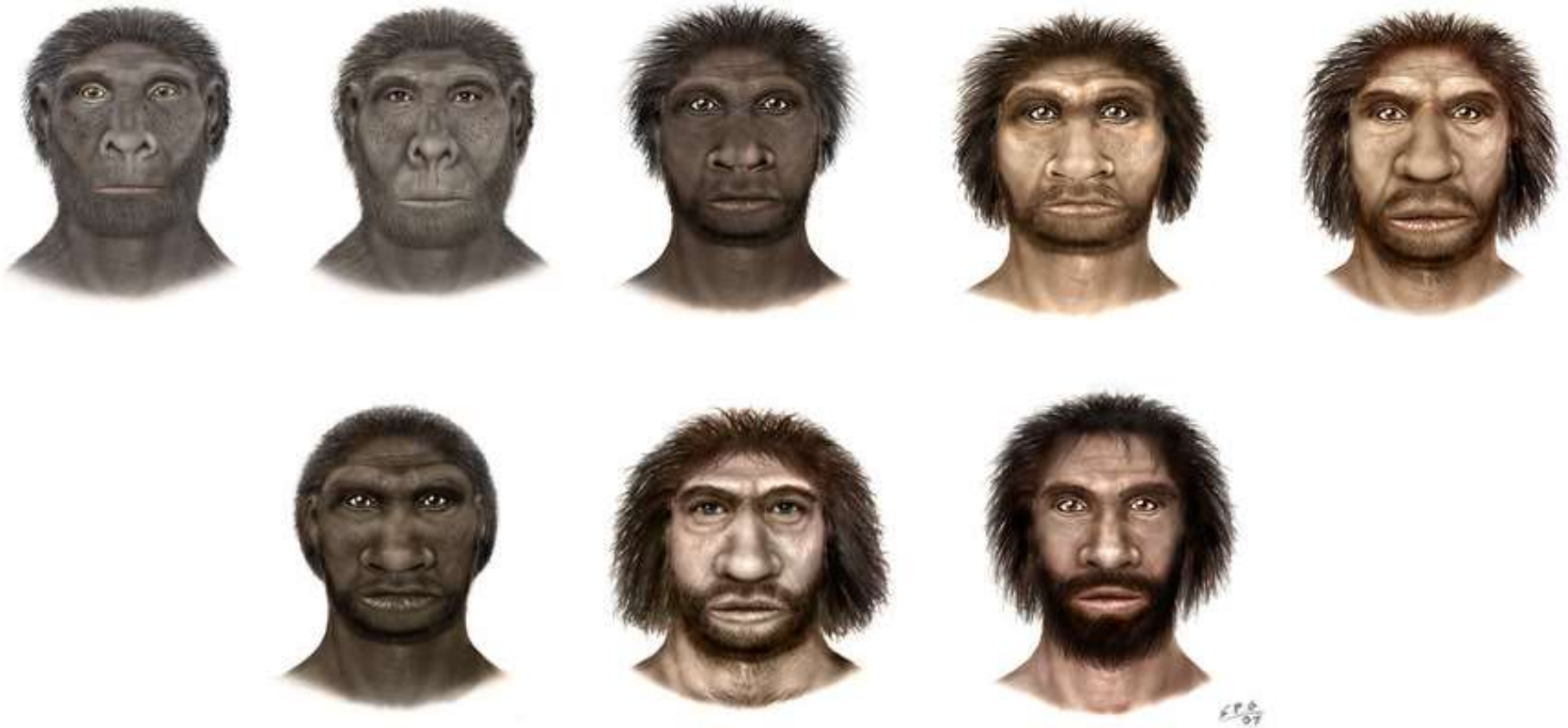


GENUS

Homo

(hominids with large brain cases, or skulls)

Meet your human taxonomy



-*Homo habilis* (have long hands)-*H. rudolfensis* -*H. ergaster* (African upright man) -*H. erectus* (upright man) -*H. heidelbergensis* -*H. rhodesiensis* –
H. neanderthalensis -*H. sapiens*

Meet your human taxonomy



SPECIES

H. sapiens

(the largest brain case of the genus *Homo*, giving us the capacity for complex speech; “*sapiens*” loosely translates as “knowing”)

We are the only living organisms in our species, with a unique set of combined characteristics from our family (bipedal), order (opposable thumbs), and genus (large brain case).



thank you!